

Problem

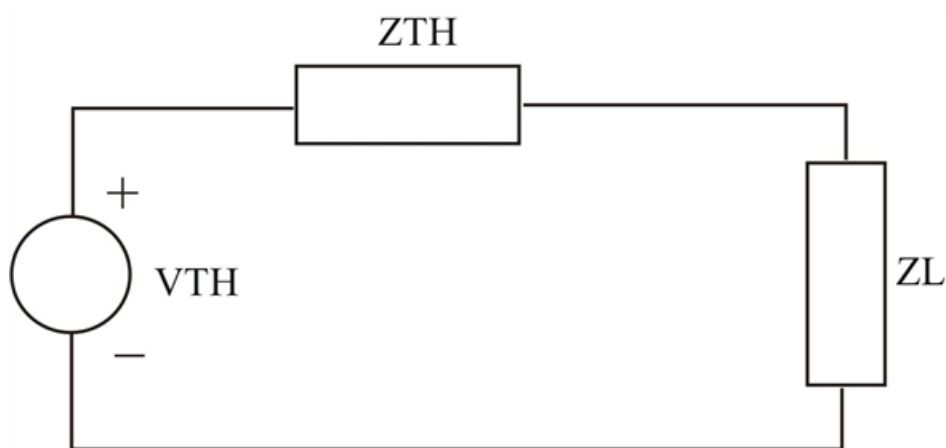
A power amplifier has an output impedance of $40 + j8 \Omega$. It produces a no-load output voltage of 146 V at 300 Hz.

(a) Determine the impedance of the load that achieves maximum power transfer.

(b) Calculate the load power under this matching condition.

Step-by-step solution

Step 1 of 3



Step 2 of 3

$$\begin{aligned} \text{(A)} \quad V_{th} &= 146V, \quad 300 \text{ Hz} \\ Z_{th} &= 40 + j8\Omega \\ Z_L &= Z_{th}^* = 40 - j8\Omega. \end{aligned}$$

Step 3 of 3

$$\text{(B)} \quad \frac{(V_{th})^2}{8R_{th}} = \frac{146^2}{(8)(40)} = 66.61 \text{ W}.$$